

# Inferential Statistics

## Testing Techniques — Concept Summary

A quick-reference guide: z-test, t-test, chi-squared, F-test/ANOVA, correlation & regression significance, and common nonparametric alternatives.

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### 1. How to Choose the Right Test

The choice of test depends on three things: (1) what type of data you have (continuous, count/categorical, ranked), (2) how many groups or variables you are comparing, and (3) whether the population standard deviation is known and whether the sample is large enough to invoke the Central Limit Theorem. The table below is a fast decision aid; each test is explained in full afterward.

| Goal                                             | Data type            | Groups      | Recommended test                                                  |
|--------------------------------------------------|----------------------|-------------|-------------------------------------------------------------------|
| Compare a sample mean to a known value           | Continuous           | 1           | One-sample z-test ( $\sigma$ known) or t-test ( $\sigma$ unknown) |
| Compare two independent group means              | Continuous           | 2           | Independent (two-sample) t-test                                   |
| Compare paired / before-after measurements       | Continuous           | 2 (paired)  | Paired t-test                                                     |
| Compare 3+ group means                           | Continuous           | 3+          | One-way ANOVA (F-test)                                            |
| Compare two population proportions               | Categorical (binary) | 2           | Two-proportion z-test                                             |
| Test observed vs. expected frequencies           | Categorical          | 1 variable  | Chi-squared goodness-of-fit                                       |
| Test association between two categorical vars    | Categorical          | 2 variables | Chi-squared test of independence                                  |
| Test a linear relationship between two variables | Continuous           | 2 vars      | Correlation / regression t-test                                   |
| Compare variances of two groups                  | Continuous           | 2           | F-test for equality of variances                                  |
| Non-normal data, 2 independent groups            | Ordinal/skewed       | 2           | Mann-Whitney U test                                               |
| Non-normal data, paired                          | Ordinal/skewed       | 2 (paired)  | Wilcoxon signed-rank test                                         |
| Non-normal data, 3+ groups                       | Ordinal/skewed       | 3+          | Kruskal-Wallis test                                               |

## 2. Z-Tests

Z-tests rely on the standard normal distribution. They are appropriate when the population standard deviation ( $\sigma$ ) is known, or when the sample size is large (commonly  $n \geq 30$ ) so the sampling distribution of the mean or proportion is approximately normal by the Central Limit Theorem.

### 2.1 One-Sample Z-Test

|                           |                                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Testing whether a sample mean differs from a known population mean, with $\sigma$ known.                                                                                  |
| <b>Hypotheses</b>         | $H_0: \mu = \mu_0$ $H_1: \mu \neq \mu_0$ (or $>$ , $<$ for one-tailed)                                                                                                    |
| <b>Test statistic</b>     | $z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$                                                                                                                             |
| <b>Degrees of freedom</b> | Not applicable (uses the normal distribution directly, no df)                                                                                                             |
| <b>Key assumptions</b>    | $\sigma$ (population std. dev.) is known; data approximately normal or $n$ is large.                                                                                      |
| <b>Example use case</b>   | A machine is calibrated to fill bottles with 500 ml, $\sigma = 5$ ml known from long-run process data. Test whether a sample of 40 bottles suggests the mean has drifted. |

### 2.2 Two-Proportion Z-Test

|                           |                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Comparing proportions (rates) between two independent groups, e.g. conversion rates, defect rates, or approval rates.           |
| <b>Hypotheses</b>         | $H_0: p_1 = p_2$ $H_1: p_1 \neq p_2$                                                                                            |
| <b>Test statistic</b>     | $z = (\hat{p}_1 - \hat{p}_2) / \sqrt{[\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)]}$ , where $\hat{p}$ is the pooled proportion           |
| <b>Degrees of freedom</b> | Not applicable                                                                                                                  |
| <b>Key assumptions</b>    | $np$ and $n(1-p) \geq 5$ for both groups; independent random samples.                                                           |
| <b>Example use case</b>   | Comparing the excavator warranty-claim rate between two dealer regions to see if one region has a genuinely higher defect rate. |

## 3. T-Tests

T-tests use the Student's t-distribution instead of the normal distribution. They are the correct choice whenever  $\sigma$  is unknown and estimated from the sample — which is the typical real-world situation, especially with small samples. As sample size grows, the t-distribution converges to the normal distribution.

### 3.1 One-Sample T-Test

|                           |                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Testing a sample mean against a known/hypothesized value when $\sigma$ is unknown.       |
| <b>Hypotheses</b>         | $H_0: \mu = \mu_0$ $H_1: \mu \neq \mu_0$                                                 |
| <b>Test statistic</b>     | $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$                                                 |
| <b>Degrees of freedom</b> | $df = n - 1$                                                                             |
| <b>Key assumptions</b>    | Population approximately normal (especially for small $n$ ); random, independent sample. |

|                         |                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Example use case</b> | Testing whether the average fuel consumption of a sampled fleet of excavators differs from the manufacturer's rated 12 L/hr, using only the sample's own std. dev. |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.2 Independent (Two-Sample) T-Test

|                           |                                                                                                                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Comparing the means of two separate, unrelated groups.                                                                                                                                                |
| <b>Hypotheses</b>         | $H_0: \mu_1 = \mu_2$ $H_1: \mu_1 \neq \mu_2$                                                                                                                                                          |
| <b>Test statistic</b>     | $t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_p^2/n_1 + s_p^2/n_2)}$ — pooled version (equal variances);<br>Welch's $t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$ if variances are unequal |
| <b>Degrees of freedom</b> | $df = n_1 + n_2 - 2$ (pooled); Welch-Satterthwaite formula for unequal variances                                                                                                                      |
| <b>Key assumptions</b>    | Both groups approximately normal; independent observations; check variance equality with an F-test or Levene's test to choose pooled vs. Welch's version.                                             |
| <b>Example use case</b>   | Comparing average bucket-cycle time between two excavator models (SK130 vs. a competitor) tested by different, unrelated operators.                                                                   |

### 3.3 Paired T-Test

|                           |                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Comparing two related measurements on the same subjects (before/after, matched pairs).                                                         |
| <b>Hypotheses</b>         | $H_0: \mu_d = 0$ $H_1: \mu_d \neq 0$ , where $d$ = difference per pair                                                                         |
| <b>Test statistic</b>     | $t = \bar{d} / (sd / \sqrt{n})$                                                                                                                |
| <b>Degrees of freedom</b> | $df = n - 1$ ( $n$ = number of pairs)                                                                                                          |
| <b>Key assumptions</b>    | The differences are approximately normally distributed; pairs are dependent by design.                                                         |
| <b>Example use case</b>   | Measuring the same excavator's fuel efficiency before and after a hydraulic system upgrade to test whether the upgrade made a real difference. |

## 4. Chi-Squared ( $\chi^2$ ) Tests

Chi-squared tests work with categorical/count data rather than means. They compare observed frequencies against what would be expected under a null hypothesis.

### 4.1 Chi-Squared Goodness-of-Fit Test

|                           |                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Testing whether a single categorical variable's observed distribution matches an expected (theoretical) distribution.           |
| <b>Hypotheses</b>         | H0: the observed distribution matches the expected distribution    H1: it does not                                              |
| <b>Test statistic</b>     | $\chi^2 = \sum (O_i - E_i)^2 / E_i$                                                                                             |
| <b>Degrees of freedom</b> | df = k - 1 (k = number of categories)                                                                                           |
| <b>Key assumptions</b>    | Expected count in each category $\geq 5$ ; independent random sample.                                                           |
| <b>Example use case</b>   | Testing whether excavator sales across five SEA countries match the split predicted by last year's regional market-share model. |

### 4.2 Chi-Squared Test of Independence

|                           |                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Testing whether two categorical variables are associated (or independent) of each other.                                                |
| <b>Hypotheses</b>         | H0: the two variables are independent    H1: they are associated                                                                        |
| <b>Test statistic</b>     | $\chi^2 = \sum \sum (O_{ij} - E_{ij})^2 / E_{ij}$ , where $E_{ij} = (\text{row total} \times \text{column total}) / \text{grand total}$ |
| <b>Degrees of freedom</b> | df = (r - 1)(c - 1) (r = rows, c = columns)                                                                                             |
| <b>Key assumptions</b>    | Expected count in each cell $\geq 5$ (as a rule of thumb); independent observations.                                                    |
| <b>Example use case</b>   | Testing whether dealer region (Thailand, Vietnam, Myanmar) is associated with customer preference for a given excavator model class.    |

## 5. ANOVA and the F-Test

### 5.1 One-Way ANOVA (Analysis of Variance)

|                           |                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Comparing means across three or more independent groups at once, avoiding the inflated Type I error from running many pairwise t-tests.                                                      |
| <b>Hypotheses</b>         | H0: $\mu_1 = \mu_2 = \dots = \mu_k$ H1: at least one mean differs                                                                                                                            |
| <b>Test statistic</b>     | $F = MS_{\text{between}} / MS_{\text{within}}$ (ratio of between-group to within-group variance)                                                                                             |
| <b>Degrees of freedom</b> | df1 = k - 1 (between groups); df2 = N - k (within groups)                                                                                                                                    |
| <b>Key assumptions</b>    | Normal populations; homogeneity of variance across groups; independent samples. A significant F is usually followed by post-hoc tests (e.g. Tukey HSD) to find which specific groups differ. |
| <b>Example use case</b>   | Comparing average dealer satisfaction scores across four different excavator model lines to see if at least one line performs differently.                                                   |

### 5.2 F-Test for Equality of Two Variances

|                           |                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Checking whether two populations have equal variances — often a preliminary step before choosing pooled vs. Welch's t-test.                     |
| <b>Hypotheses</b>         | $H_0: \sigma_1^2 = \sigma_2^2$ $H_1: \sigma_1^2 \neq \sigma_2^2$                                                                                |
| <b>Test statistic</b>     | $F = s_1^2 / s_2^2$ (larger sample variance in the numerator by convention)                                                                     |
| <b>Degrees of freedom</b> | $df_1 = n_1 - 1$ , $df_2 = n_2 - 1$                                                                                                             |
| <b>Key assumptions</b>    | Both populations approximately normal — the F-test is quite sensitive to non-normality; Levene's test is a more robust alternative in practice. |
| <b>Example use case</b>   | Checking whether cycle-time variability differs between two excavator operators before comparing their average cycle times.                     |

## 6. Correlation and Regression Significance

### 6.1 T-Test for Pearson Correlation / Regression Slope

|                           |                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Used when</b>          | Testing whether a linear relationship between two continuous variables is statistically different from zero.                                |
| <b>Hypotheses</b>         | $H_0: \rho = 0$ (or $\beta = 0$ for a regression slope) $H_1: \rho \neq 0$                                                                  |
| <b>Test statistic</b>     | $t = r\sqrt{(n-2) / \sqrt{1-r^2}}$ (for correlation); analogous form using the coefficient's standard error for a regression slope          |
| <b>Degrees of freedom</b> | $df = n - 2$                                                                                                                                |
| <b>Key assumptions</b>    | Linear relationship; bivariate normality (for correlation); homoscedastic, independent residuals (for regression).                          |
| <b>Example use case</b>   | Testing whether the correlation between machine operating hours and maintenance cost is significantly different from zero across the fleet. |

## 7. Nonparametric Alternatives

Nonparametric tests avoid the normality assumption and work on ranks rather than raw values, making them useful for small samples, skewed data, or ordinal scales.

| Test                        | Parametric equivalent         | Use case                                                                  |
|-----------------------------|-------------------------------|---------------------------------------------------------------------------|
| Mann-Whitney U              | Independent two-sample t-test | Compare two independent groups when data are skewed or ordinal            |
| Wilcoxon signed-rank        | Paired t-test                 | Compare paired/matched samples when differences aren't normal             |
| Kruskal-Wallis              | One-way ANOVA                 | Compare 3+ independent groups when normality/equal variance fail          |
| Spearman's rank correlation | Pearson correlation           | Test monotonic (not necessarily linear) association between two variables |

## 8. Reading the Result: p-value, $\alpha$ , and Errors

- p-value: the probability of observing a result at least as extreme as the sample, assuming  $H_0$  is true. Reject  $H_0$  if p-value < significance level ( $\alpha$ , commonly 0.05).
- Type I error ( $\alpha$ ): rejecting a true  $H_0$  (false positive). Type II error ( $\beta$ ): failing to reject a false  $H_0$  (false negative). Power =  $1 - \beta$ .
- One-tailed vs. two-tailed: use one-tailed only when the direction of the effect is specified in advance by the research question; otherwise use two-tailed.
- Statistical vs. practical significance: with a large enough sample, even a tiny, practically meaningless difference can become statistically significant — always check effect size alongside the p-value.

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Quick reference summary — verify formulas and assumptions against your course textbook/notes before an exam.